



The  $3 \times 3$  matrix  $\mathbf{A}$  has distinct non-zero eigenvalues  $a, \frac{1}{2}, 2$  with corresponding eigenvectors

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -1 \\ 2 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix},$$

respectively.

(c) Find  $\mathbf{A}^{-1}$  in terms of  $a$ .

[5]

[illegible]